MILITARY SPECIFICATION

FUZE, XM596; SETBACK PIN AND HOUSING ASSEMBLY FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the metal parts and loading, assembling and packing for one type of setback pin and housing assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental, and Performance Tests For

FSC: 1390

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8880916 - Setback Pin and Housing Assembly

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-8880916 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all requirements specified on drawing (dwg.) 8880916, all associated drawings and with all requirements specified in applicable specifications.

3.3 Transportation-vibration.-The setback pin and housing assembly shall be safe and operable after test specified in 4.4.11.

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3.4 Jolt and Jumble.-The setback pin and housing assembly shall withstand the jolt and jumble tests without having any explosive element explode, assembly come apart, assembly arm, any safety device fails or become hazardous to anyone handling them when tested as specified in 4.4.10.

3.5 Functioning.-The setback pin and housing assembly shall function satisfactorily when tested as specified in 4.4.12.

3.6 Non-functioning.-The setback pin and housing assembly shall not function when tested as specified in 4.4.13.

3.7 Workmanship.-All parts and assemblies shall be fabricated, finished, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.8 Submission of results of contractor-conducted examinations and test.-The Contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.6).

3.9 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Three hundred thirty-four (334) setback pin and housing assemblies.
b. One hundred ten (110) sub-assemblies.
c. Fifty (50) each of every component.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. The first article quantity, or portion thereof, as directed by the Contracting

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Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.2.1 Quantities for first article inspection.-After classification of defects and other inspections have been performed as specified in 4.2.2, the following quantities shall be subjected to the indicated tests:

4.2.2.1.1 Qualitative spot test for tin.-Five (5) rotor springs and ten (10) lock springs shall be subjected to this test.

4.2.2.1.2 Stake of pinion assembly.-The one hundred ten (110) pinion assemblies shall be subjected to this test.

4.2.2.1.3 Stake of verge assembly.-The one hundred ten (110) verge assemblies shall be subjected to this test.

4.2.2.1.4 Qualitative spot test for cadmium (when applicable).-Five (5) rotor springs shall be subjected to this test.

4.2.2.1.5 Arming delay time and non-arming of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.6 Non-arming of centrifugal lock of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.7 Arming of centrifugal lock of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.8 Push test of simulated setback pin into retainer setback pin.-The fifty (50) retainer setback pins shall be subjected to this test.

4.2.2.1.9 Jolt.-Twelve (12) setback pin and housing assemblies shall be subjected to this test.

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4.2.2.1.10 Jumble.-Twelve (12) setback pin and housing assemblies shall be subjected to this test.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Remaining first article inspection samples.

4.2.4.1 Transportation vibration.-Three hundred (300) setback pin and housing assemblies, packed in their regular shipping containers, shall be subjected to this test.

4.2.4.2 Functioning.-After completion of tests specified in 4.2.4.1, the three hundred (300) assemblies shall be subjected to this test. If eight (8) or more assemblies fail to comply with the applicable requirements, first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4.3 Non-functioning.-Ten (10) setback pin and housing assemblies shall be subjected to this test. If any assembly functions, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

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4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Detonators from one interfix lot number from one manufacturer.
b. Metal parts from one interfix lot number from one manufacturer which will include retainers from one heat treated batch.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Pin, setback (see dwg. 8876211 covering a detail of dwg. 8880916)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Diameter of shaft at end toward retainer	Gage	01001
102.	Length of small diameter	Gage	01002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	01003

4.3.2.2 Retainer, setback pin (see dwg. 8876212 covering a detail of dwg. 8880916)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Depth	Gage	02001
102.	Diameter, retaining mounds	Gage	02002
103.	Concentricity retaining mounds diameter with center diameter	Gage	02003
104.	Center diameter	Gage	02004
Minor:	AQL 1.00 Percent		
201.	Any retaining mounds missing	Visual	02005
202.	Evidence of poor workmanship (see 3.7)	Visual	02006

4.3.2.3 Plate, top (see dwg. 8876194 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Outside diameter	Gage	03001
102.	Location of dowels	Gage	03002
103.	Height of dowels	Gage	03003

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03004

4.3.2.4 Plate, shaft, lower (see dwg. 8876199 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Outside diameter	Gage	04001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	04002

4.3.2.5 Plate, spacer, lower (see dwg. 8876198 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Outside diameter	Gage	05001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05002

4.3.2.6 Plate, shaft, upper (see dwg. 8876195 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Outside diameter	Gage	06001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	06002

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4.3.2.7 Plate, spacer, upper (see dwg. 8876196 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Outside diameter	Gage	07001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	07002

4.3.2.8 Plate, rotor (see dwg. 8876197 covering a detail of dwg. 8880915)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Outside diameter	Gage	08001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	08002

4.3.2.9 Rotor and gear assembly (see dwg. 8876202 covering a detail of dwg. 8876220)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Rotor, weight incorrect and coining missing	Visual	09001
Minor:	None defined		

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4.3.2.10 Setback pin and housing assembly (see dwg. 8880916)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor fails to remain in unarmed position when centrifugal lock is retracted	Visual/Manual	10001
2.	Setback pin not through rotor gear	Probe	10002
3.	Setback pin retainer missing	Visual/Probe	10003
4.	Two (2) or more leaves of setback pin retainer will not contact setback pin (see 4.3.2.11.1)	Visual	10004
Major:	AQL 0.40 Percent		
101.	One (1) leaf will not contact setback pin (see 4.3.2.11.1)	Visual	10005
102.	Discernible bulge on detonator	Visual	10006
103.	Total height	Gage	10007

Minor: None defined

4.3.2.11 Defect standards - Code No. 11001

4.3.2.11.1 Examination for contact of setback pin retainer leaves (see para. 4.3.2.10).-Each assembly in which one (1) or more leaves of the setback pin retainer apparently will fail to contact the setback pin shall be subjected to the following:

a. The setback pin shall be operated.
b. Visual examination shall be conducted to determine whether leaves contact the setback pin.

If two (2) or more leaves fail to contact the setback pin, the assembly shall be classed a critical defective. If one (1) leaf fails to contact the setback pin, the assembly shall be classed a Major defective. Assemblies subjected to examination in accordance with this procedure shall not be returned to the lot.

4.3.3 Testing

4.3.3.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (1) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In

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addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.2 Check test for deterioration of detonators (see applicable detonator specification), Major Defect.-If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in the applicable detonator specification immediately before the detonator lot is assembled into the fuze. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembly the detonator into the fuze. (See 6.4)

4.3.3.3 Qualitative spot test for tin (see dwgs. 8876214 and 8876215), Major Defect, Code No. 12001.-Ten (10) lock springs and five (5) rotor spings shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement, the lot shall be rejected.

NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one roll of spring wire.

This test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

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4.3.3.4 Qualitative spot test for cadmium (see dwg. 8876215), when applicable, Major Defect, Code No. 13001).-Five (5) rotor springs shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement, the lot shall be rejected.

NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one roll of spring wire. This test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.5 Stake of pinion assembly (see dwg. 8876210), Major Defect, Code No. 14001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.6 Stake of verge assembly (see dwg. 8876207), Major Defect, Code No. 15001.-The sample plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.7 Arming delay time (see dwg. 8876220), Critical Defect if assembly arms in less than 100 milliseconds, Major Defect if assembly arms between 100 and 140 milliseconds - Non-arming of housing assembly, Major Defect, Code No. 16001.-This test shall be conducted 100 percent. If any housing assembly fails to comply with the time limits specified on the applicable drawing, it shall be classified defective and removed from the lot. This test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.8 Push test of simulated setback pin into retainer, setback pin (see dwg. 8876212), Major Defect, Code No. 17001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be perfromed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.9 Non-arming of centrifugal lock of housing assembly (see dwg. 8876220), Major Defect, Code No. 18001.-This test shall be conducted 100 percent. If any housing assembly fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot. This test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

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4.3.3.10 Arming of centrifugal lock of housing assembly (see dwg. 8876220), Major Defect, Code No. 19001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter L, single plan, with an AQL of 0.40 percent. This test shall be conducted after completion of test specified in 4.3.3.9. This test shall be performed as specified in 4.4.8 using equipment specified in 4.3.4.

4.3.3.11 Transportation vibration test of centrifugal lock mechanism (see dwg. 8876220), Major Defect, Code No. 20001.-Twenty (20) unarmed assemblies without setback pins and without detonators, shall be selected from each lot assemblied to M533 fuzes and subjected to this test. Each assembly shall be conspicuously marked to indicate absence of setback pin and detonator. Assemblies shall be placed in interplant shipping containers and subjected to this test.

At the outset of the test, the rotor shall be in the fully unarmed position, with the rotor gear in contact with a pin with a diameter of .078 - .001 inches which passes through the rotor gear and rotor plate. The centrifugal lock shall be in engagement with the star wheel. The pin shall then be removed and the fuze assembly shall be subjected to the test. The amount of travel of each rotor shall be noted. If the travel of any rotor exceeds 26 degrees the lot shall be rejected. All data concerning rotor travel shall be reported to Picatinny Arsenal, ATTN: SMUPA-ND3. This test shall be performed as specified in 4.4.9 using equipment specified in 4.3.4.

4.3.3.12 Jolt (see 3.4), Major Defect, Code No. 21001.-Twelve (12) setback pin and housing assemblies shall be selected from each lot, assembled to M533 fuzes, and subjected to this test. If any assembly fails to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.10 using equipment specified in 4.3.4.

4.3.3.13 Jumble (see 3.4), Major Defect, Code No. 22001.-Ninety six (96) setback pin and housing assemblies shall be selected from each lot, assembled to M533 fuzes, and subjected to this test. If any assembly fails to comply with the applicable requirement the lot shall be rejected. This test shall be performed as specified in 4.4.10 using equipment specified in 4.3.4.

4.3.3.14 Transportation Vibration (see 3.3), Major Defect, Code No. 23001.-Two-hundred and ten (210) setback pin and housing assemblies shall be selected from each lot, placed in their regular shipping containers and subjected to this test. This test shall be performed as specified in 4.4.11 using equipment specified in 4.3.4.

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4.3.3.15 Functioning (see 3.6), Major Defect, Code No. 24001.-After completion of test specified in 4.3.3.14, two hundred (200) assemblies shall be subjected to this test. If eight (8) or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.12 using equipment specified in 4.3.4.

4.3.3.16 Non-functioning (see 3.7), Major Defect, Code No. 25001.-Ten (10) additional assemblies shall be selected from each lot and subjected to this test. If any assembly functions, the lot shall be rejected. This test shall be performed as specified in 4.4.13 using equipment specified in 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8880916 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Qualitative spot test for tin.-The rotor springs and lock springs shall be tested in accordance with the following procedure:

a. Prepare 50 ml. of saturated solution of cacotheline by dissolving small quantities of cacotheline powder in water. The solution can be considered saturated when the addition of small amounts of cacotheline will no longer dissolve.

b. Clean the parts in acetone and place 2 lock springs or one (1) rotor spring in a clean test tube. Add 10 drops of concentrated hydrochloric acid and allow tin to dissolve for 10 minutes. Then add 5 drops of distilled water and mix well.

c. Moisten No. 42 Whatman filter paper with saturated cacotheline solution and allow to partially dry.

d. Place a drop of test solution from the test tube on damp filter papter. The development of a distinct red-violet spot where the test solution was placed indicates the presence of tin.

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This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Qualitative spot test for cadmium.-The rotor springs shall be tested in accordance with the following procedure:

a. Clean the parts in acetone and place one (1) rotor spring in a clean test tube. Add 10 ml. concentrated hydrochloric acid and allow cadmium to dissolve.

b. Decant fluid into a beaker and dilute with distilled water to total 100 ml.

c. Add 50 ml. saturated hydrogen sulfide.

d. The development of a yellow precipitant indicates the presence of cadmium.

This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Stake of pinion assembly.-The pinion assembly shall be placed in a fixture specified on the Equipment List and the torque specified on dwg. 8876210 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.4 Stake of verge assembly.-The verge assembly shall be placed in a fixture specified on the Equipment List and the load specified on dwg. 8876207 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.5 Arming delay time and non-arming of housing assembly.-An electronic time device and an electrically operated retractable simulated setback pin shall be operated simultaneously. A light source shall be concentrated on the detonator hole in the rotor. A photoelectric cell shall not trip an electronic circuit before the non-arming time specified. For the arming time, the photoelectric cell shall trip an electric circuit by the arming time specified. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.6 Push test of simulated setback pin into retainer setback pin.-The retainer setback pin shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8876212 shall be applied to a simulated mean setback pin. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.7 Non-arming of centrifugal lock of housing assembly, Method 1.-The housing assembly shall be mounted in a centrifuge and spun about its axis of symmetry at the rotational speed specified for a minimum of 0.400 seconds. Visually check to see

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that the rotor has not moved from the unarmed position. When the visual check provides doubtful results, a gage shall be used which simulated the minimum diameter set back pin. An unarmed housing assembly shall be capable of accepting the simulated pin through the rotor plate and rotor gear.

Method 2.-The housing shall be mounted and spun as in Method 1. Upon completion of spin, a light source shall be concentrated on the rotor through the setback pin check hole in the top plate. A photoelectric cell shall trip an electric circuit. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.8 Arming of centrifugal lock of housing assembly, Method 1.-The housing assembly shall be mounted in a centrifuge and spun about its axis of symmetry at the rotational speed specified. Visually check to see that the rotor moves from the unarmed position to the armed position.

Method 2.-The housing shall be mounted and spun as in Method 1. Upon completion of spin, a light source shall be concentrated on the rotor through the setback pin check hole in the top plate. A photoelectric cell shall trip an electric circuit. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Transportation vibration test of centrifugal lock mechanism.-The mechanism shall be tested as specified in MIL-STD-331. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.10 Jolt and Jumble.-The setback pin and housing assemblies shall be tested in accordance with the procedures specified in MIL-STD-331. These tests are destructive tests. Parts so tested shall not be returned to the lot.

4.4.11 Transportation vibration.-The setback pin and housing assemblies shall be tested in accordance with MIL-STD-331. Parts subjected to this test shall not be returned to the lot but may be used for subsequent tests.

4.4.12 Functioning.-The setback pin and housing assemblies shall be assembled to a M533 fuze, assembled to a test cartridge (dwg. 9206914) and fired from a launcher for which it is standard to impact at 200 meters plus or minus 5 meters onto coarse sand, ten inches minimum thickness to point of impact. The angle of firing shall be adjusted to give an impact of approximately 30 degrees. Rake sand after every round or traverse weapon between shots so that impacts are not overlapping. If a round falls outside the specified area, it shall be declared a "no test" round. When results for functioning are reported, the number of "no test" rounds shall also be reported.

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4.4.13 Non-functioning.-The setback pin and housing assembly shall be assembled to a M533 fuze, assemble to a test cartridge (dwg. 9206914) and fired from a launcher for which it is standard against a vertical celotex target placed 60 feet from the launcher. Observation shall be made for non-functioning.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The setback pin and housing assembly shall be preserved and packaged in accordance with dwg. 8880918.

5.2 Packing

5.2.1 Level C.-The setback pin and housing assembly, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with dwg. 9230368.

5.3 Marking.-Marking shall be in accordance with dwg. 8799112 and Code of Federal Regulations, Title 49, Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor designs on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally exploys in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility

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and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonator assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonator assemblies, or to the contractor assembling the detonator assemblies into the setback pin and housing assembly; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator assembly lot was under the jurisdiction of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the assembly is procured by the U.S. Army Munitions Command one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
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Preparing activity:
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